

VALENTIN VELEV

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Frankfurt am Main, Germany

EXPERIENCE

Working Student, Data Science

March 2025 – August 2026

Codify AG

Kreuzlingen, Switzerland

- Designed and built an end-to-end evaluation pipeline for RAG and agentic AI systems, enabling reproducible benchmarking of across use cases, using Python, SQL, DeepEval, MCP, and Arize Phoenix
- Scaled LLM-powered evaluation workflows across provider APIs, balancing token consumption, latency, and cost for large evaluation runs
- Standardised benchmarking across multiple AI use cases and built a framework for evaluating multi-turn conversations
- Integrated end-to-end evaluation pipelines and interactive visualisations into an existing GCP-based production environment
- Designed hands-on prompt-injection challenges for a workshop at the AIXCHANGE Conference 2025, translating common LLM failure modes into practical exercises on robustness and responsible AI use

Teaching Assistant

August 2025 – February 2026

Department of Computer and Information Science, University of Konstanz

Konstanz, Germany

- Taught the master's course "Word Representations and Language Models", covering embedding models, sequence-to-sequence models, the Transformer architecture (incl. BERT) and generative language models
- Developed and maintained reproducible teaching materials using Python, Git, and $\LaTeX$

Research Assistant

April 2024 – February 2025

Center for Data and Methods, University of Konstanz

Konstanz, Germany

- Conducted independent quantitative analyses and contributed to several peer-reviewed publications
- Built reproducible analysis pipelines, statistical models and data visualisations in R, Python and Stata

EDUCATION

M.Sc. Data Science (Social and Economic Data Science)

October 2023 – March 2026

University of Konstanz

Konstanz, Germany

- Final grade: 1.1 (with distinction)
- Coursework: Deep Learning, Language Modelling, Statistical Learning, Data Mining
- Master's thesis on cross-domain evaluation of retrieval models

B.A. Political Science

October 2019 – January 2023

Goethe University Frankfurt

Frankfurt am Main, Germany

- Final grade: 1.2 (with distinction)
- Coursework: Statistics, Data Science for Human Geography

PROJECTS

gglyph | R | [CRAN](#), [GitHub](#)

September 2025

- Built, documented, published, and maintained an open-source R package on CRAN for visualising directed pairwise relationships
- Set up a CI/CD pipeline with GitHub Actions (cross-platform R CMD checks, test coverage, automated documentation)

EfficientPQM | Rust, Python

Ongoing

- Implementing performant metrics for evaluating dimensionality-reduction methods in Rust with Python bindings
- Currently adding GPU and Apple MPS support for further acceleration

Cross-Domain Evaluation of Retrieval Models | Python | [GitHub](#)

Master's Thesis

- Reproducible evaluation of document retrieval models across multiple domains
- Systematic error analysis comparing model architectures and evaluation metrics

CERTIFICATIONS

IBM DevOps and Software Engineering Professional Certificate

Since August 2026 (in progress)

IBM / Coursera

Online

- *DevOps & cloud native*: CI/CD (GitHub Actions, Jenkins), Infrastructure as Code (Terraform), Docker, Kubernetes, OpenShift, microservices, serverless
- *Software engineering*: Agile/Scrum, TDD/BDD and unit testing, REST APIs (Flask), Linux/Bash scripting, monitoring and observability (Prometheus, Grafana)

IBM Data Engineering Professional Certificate

Since August 2026 (in progress)

IBM / Coursera

Online

- *Pipelines & ETL*: Apache Airflow, Kafka, Spark/PySpark, Hadoop, Bash
- *Databases & warehousing*: PostgreSQL, MySQL, IBM Db2, NoSQL (MongoDB, Cassandra), data warehouse design (star/snowflake schema)

SKILLS

Technical skills:

Programming: Python, SQL, R, Rust, Java, HTML/CSS

AI & NLP: PyTorch, Transformers, RAG, agentic AI, LLM evaluation (DeepEval, Arize Phoenix), prompt engineering, fine-tuning, information retrieval, MCP, LangChain, LlamaIndex, ChromaDB

Engineering: Git, Docker, Kubernetes, CI/CD, REST APIs, Agile

Cloud: GCP, ETL, Apache Airflow, Kafka, Spark, Data Warehousing, PostgreSQL, NoSQL (MongoDB)

Analysis & Visualisation: statistical modelling, benchmarking, experimental design, pandas, matplotlib, Tableau, Stata, Data Studio, $\LaTeX$

Languages: German and Bulgarian – native; English – C2; Dutch – A1

PUBLICATIONS

Velev, V., & Spitz, A. (2026). [A Systematic Multi-Domain Evaluation of Document Retrievers](#). arXiv:2609.29455 [cs.IR].

Velev, V., Fuchs, J., & Spitz, A. (2026). [An R Package for Visualizing Significance in Directed Pairwise Relationships Using Tapered Edge Glyphs](#). *Proceedings of the 28th Eurographics Conference on Visualization (EuroVis 2026) – Posters and Demos*.

Velev, V., & Schmidt-Catran, A. W. (2024). [Economic Inequality and Public Support for Redistribution in Europe: A Cross-Sectional and Longitudinal Multilevel Analysis](#). *International Journal of Public Opinion Research*, 36(4), edae055.